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Injection molding apparatus with load-balancing mechanism

Abstract

An apparatus for injecting a molten material including an injector, a rail, a bell crank, and a frame supporting the injector via the rail for guiding the linear translation of the injector along the frame. The apparatus includes a first guide and a second guide slideably coupled to the rail, the second guide attached to the injector and spaced from the first guide along the rail, and a bell crank coupled to the first guide and second guide to provide a load-balancing mechanism.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 17/317,288, filed May 11, 2021, the entire contents of which are incorporated herein by reference.

FIELD

(1) The invention relates generally to an injection molding apparatus, and in particular, to an injection molding apparatus with an injector coupled to a load-balancing mechanism.

BACKGROUND

(2) In some injection molding apparatuses, an injector, with a nozzle at an end, is advanced by an actuator towards an inlet of a mold. Once the nozzle is in contact with the inlet of the mold, the actuator urges the nozzle against the mold to prevent the nozzle from separating from the mold as the injector injects molten material into the mold. If the force applied by the actuator is not aligned with the force experienced by the injector, the nozzle of the injector can misalign with the inlet of the mold.

BRIEF SUMMARY

- (3) An aspect of the present application provides an apparatus for injecting a molten material comprising: an injector for injecting the molten material; a frame supporting the injector; a rail for guiding the linear translation of the injector along the frame, the rail supported by the frame; a first guide and a second guide slideably coupled to the rail, the second guide attached to the injector and spaced from the first guide along the rail; and a bell crank including a first location, a second location, and a third location arranged in a triangle, the bell crank rotatably coupled to the first guide at the first location, the bell crank rotatably coupled to the second guide at the second location such that applying a linear force at the third location in a first direction from the second guide towards the first guide translates the first and second guides in the first direction along the rail.
- (4) The rail can include two lateral opposite sides and the apparatus can further comprise an another bell crank, each of the bell crank and the another bell crank is located on a respective one of the two lateral opposite sides of the rail; the another bell crank including a fourth location, a fifth location, and a sixth location arranged in a triangle, the another bell crank rotatably coupled to the first guide at the fourth location, the another bell crank rotatably coupled to the second guide at the fifth location such that applying a linear force at the sixth location in the first direction translates the first and second guides in the first direction along the rail.
- (5) At any location along the rail, the first guide can be substantially restricted from distancing from the rail in a second direction along a line not substantially parallel to the rail.
- (6) The rail can include two longitudinal protrusions, one on each respective lateral sides of the rail, the first guide includes a body having two flanges defining a space for partially receiving the rail, each flange adjacent a respective lateral sides of the rail, each flange including a protrusion engaging a respective one of longitudinal protrusions of the rail thereby substantially restricting the first guide from distancing from the rail in the second direction.
- (7) The apparatus can comprise a third guide slideably coupled to the rail, upstream of the second guide, supporting and attached to the injector.
- (8) The bell crank can be rotatably coupled to the first guide at the first location via a pin joint and the another bell crank can be rotatably coupled to the first guide at the fourth location via a pin joint.
- (9) The bell crank can be rotatably coupled to the second guide at the second location via a pin joint and the another bell crank is rotatably coupled to the second guide at the fifth location via a pin joint.
- (10) The rail can be a linear slide rail.
- (11) The first guide can include a carriage block.
- (12) The second guide can be a trunnion including two pins one on each opposing sides of the trunnion extending away from the rail.
- (13) The apparatus can comprise a linear actuator rotatably coupled to the bell crank at the third location via a pin joint and rotatably coupled to the another bell crank at the sixth location via pin joint to translate the injector along the rail.
- (14) The linear actuator can include a motor.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The drawings are not to scale.
- (2) FIG. 1 is a side view of an apparatus for injecting a molten material according to an embodiment of the present application.
- (3) FIG. 2 is a side view of some components of the apparatus of FIG. 1 isolated from the frame.
- (4) FIG. 3 is a bottom perspective view of some components of the apparatus of FIG. 1 isolated

from the frame.

(5) FIG. 4 is a magnified view of a portion of FIG. 3.

(6) FIG. 5 is a magnified view of a portion of FIG. 3 showing a side opposite of the view shown in FIG. 4.

(7) FIG. 6 is a bottom perspective view of some components of the apparatus of FIG. 1 isolated from the frame and other components.

DETAILED DESCRIPTION

(8) Specific embodiments of the present application are now described with reference to the figures. The following detailed description is merely exemplary in nature and is not intended to limit the concepts and uses of the concepts. Furthermore, there is no intention to be restricted by any expressed or implied theory in the present application. In the description, “downstream” is used with reference to the direction of the molten material flow from an injector to a mold, and also with reference to the order of components, or features thereof, through which the molten material flows from the injector to the mold, whereas “upstream” is used with reference to the opposite direction. Depending on context, upstream and downstream can be used as relative or absolute terms.

(9) Referring to FIG. 1, an apparatus for injecting a molten material (e.g., plastic) is generally indicated by **10**. Apparatus **10** comprises a frame **15**, an injector **20**, a rail **25**, and an actuator **30** (see FIG. 2). Rail **25** is supported by frame **15** and injector **20** is supported by rail **25**. In other words, injector **20** is supported by frame **15** via rail **25**. Actuator **30** is a linear actuator coupled to injector **20** to translate injector **20** relative to frame **15** towards, or away from, an inlet **35** of a mold **40** (both shown schematically in FIG. 1). Injector **20** is for injecting a molten material (not shown) into mold **40** via inlet **35**. Rail **25** is for guiding the linear translation of injector **20** along frame **15**. (10) Referring to FIG. 2, FIG. 3, and FIG. 4, apparatus **10** comprises a first guide **45** and a second guide **50** slideably coupled to rail **25**. Second guide **50** is attached to injector **20**, spaced from first guide **45** along rail **25**, and upstream of first guide **45**. Apparatus **10** comprises a bell crank **55** including a first location **60**, a second location **65**, and a third location **70** arranged in a triangle (see FIG. 2). (Bell crank **55** is shown in phantom in some figures to show components obscured by bell crank **55**.) Bell crank **55** is rotatably coupled to first guide **45** at first location **60** and rotatably coupled to second guide **50** at second location **65** such that applying a linear force at third location **70** in a first direction A from second guide **50** towards first guide **45** (i.e., downstream direction) translates first and second guides **45**, **50** in first direction A.

(11) Referring to FIG. 5, apparatus **10** comprises a bell crank **75** (shown in phantom in FIG. 5 to show components obscured by bell crank **75**). Each of bell crank **55** and bell crank **75** is located on a respective one of lateral opposite sides **80**, **85** of rail **25**. Bell crank **75** includes a fourth location **90**, a fifth location **95**, and a sixth location **100** arranged in a triangle. Bell crank **75** is rotatably coupled to first guide **45** at fourth location **90** and rotatably coupled to second guide **50** at fifth location **95** such that applying a linear force at sixth location **100** in first direction A translates first and second guides **45**, **50** in first direction A.

(12) First guide **45** is substantially restricted from decoupling from rail **25** at any location along rail **25**. In the illustrated embodiments, first guide **45** includes a carriage block **105** and a block **110** (see FIG. 6) supported by carriage block **105** and rail **25** is a linear slide rail. Rail **25** includes two longitudinal protrusions **115**, **120**, on lateral sides **80**, **85** of rail **25**, respectively. First guide **45** includes a body **125** having two flanges **130**, **135** defining a space **140** to partially receive rail **25** (see FIG. 6). Each of flanges **130**, **135** is adjacent to one of respective lateral sides **80**, **85** of rail **25**. Each of flanges **130**, **135** includes a protrusion **145** extending into space **140** and engaging a respective one of protrusions **115**, **120** coupling first guide **45** to rail **25** thereby allowing first guide **45** to decouple from rail **25** only at any of the two ends **150**, **155** of rail **25** (see FIG. 3). That is, in the illustrated embodiments, first guide **45** is directly coupled to rail **25** via the engagement of protrusions **145** with respective protrusions **115**, **120**. In the illustrated embodiments, first guide **45** includes a protrusion **156** (implemented as a pin) on each of opposite lateral sides of first guide **45**

extending towards one of respective bell cranks **55**, **75** (see FIG. 5 and FIG. 6). In the illustrated embodiments, protrusions **156** is attached to block **110** but a person of ordinary skills in the relevant art would appreciate that protrusions **156** can be attached to carriage block **105** and block **110** can be omitted.

(13) Referring to FIG. 4 and FIG. 6, apparatus **10** comprises guides **157**, **158** supporting injector **20** on rail **25**. Guide **157** is spaced from and upstream of second guide **50**. Guide **158** is upstream of guide **157**. In the illustrated embodiments, guides **157**, **158** are carriage blocks attached to injector **20** and include protrusions (not shown) that are similar to protrusions **145** of first guide **45**. The protrusions of guides **157**, **158** engage respective protrusions **115**, **120** of rail **25** similar to the engagement of protrusions **145** with respective protrusions **115**, **120**. Unlike first guide **45**, second guide **50** is not directly coupled to rail **25** and is indirectly coupled to rail **25** via the attachment of second guide **50** to injector **20** and the attachment of injector **20** to guides **157**, **158**. In the illustrated embodiments, second guide **50** is a trunnion **160** including a body **165** with two flanges **170**, **175** defining a space **180** to partially receive rail **25** (see FIG. 6). Trunnion **160** is attached to injector **20** via bolts **182**. Each flange **170**, **175** is adjacent one of respective lateral sides **80**, **85** of rail **25**. Flange **170** includes a protrusion **185** extending away from space **180**. Flange **175** includes a protrusion **190** extending away from space **180**. In the illustrated embodiments, protrusions **185**, **190** are pins. However, a person of ordinary skill in the art would appreciate that protrusions **185**, **190** can be located at other suitable locations not on flanges **170**, **175**.

(14) In the illustrated embodiments, bell crank **55** is rotatably coupled to first guide **45** at first location **60** via a pin joint **192** and to second guide **50** at second location **65** via a pin joint **195** (i.e., protrusion **156** is partially received in bell crank **55** at first location **60** and protrusion **185** is partially received in bell crank **55** at second location **65**). In the illustrated embodiments, bell crank **75** is rotatably coupled to first guide **45** at fourth location **90** via a pin joint **197** and to second guide **50** at fifth location **95** via a pin joint **200** (i.e., protrusion **156** is partially received in bell crank **55** at fourth location **90** and protrusion **190** is partially received in bell crank **75** at fifth location **95**).

(15) Referring to FIG. 2, actuator **30** is coupled to bell crank **55** at third location **70** and to bell crank **75** at sixth location **100** (see FIG. 5). In illustrated embodiments, actuator **30** is a linear actuator and includes a motor **205**. Injector **20** comprises a barrel **210** and a nozzle **215**, which is in fluid communication with barrel **210**, to inject a molten material (not shown) into mold **40** via inlet **35** (see FIG. 1). When actuator **30** applies a force in first direction A at third location **70** and sixth location **100**, especially when nozzle **215** is urged against inlet **35** of mold **40**, because the force is offset by a perpendicular distance C from line D-D (i.e., perpendicular distance between lines D-D and E-E), barrel **210** will tend to rotate in angular direction B about location **220**. As described in the present application, actuator **30** and injector **20** are configured to convert the linear force provided by actuator **30** into a linear component to linearly translate injector **20** along rail **25** and when actuator **30** is directing the linear force in first direction A, part of that force is also converted to counter the tendency of barrel **210** to rotate nozzle **215** away from rail **25**. Reducing the tendency of barrel **210** to rotate in angular direction B involves selecting a suitable ratio between distance B, the perpendicular distance between lines B-B and C-C, and distance A, the perpendicular distance between lines A-A and D-D, and the ratio between distance D, the perpendicular distance between lines F-F and B-B, and distance C, the perpendicular distance between lines E-E and D-D.

(16) While various embodiments according to the present application have been described above, it should be understood that they have been presented by way of illustration and example only, and not limitation. It will be apparent to persons of relevant ordinary skill in the relevant art that various changes in form and detail can be made therein without departing from the scope of the present application. For example, in the illustrated embodiments, rail **25** is a separate member attached to frame **15** via bolts but a person of ordinary skills in the relevant art would appreciate that rail **25** and frame **15** can be implemented as a unitary member. It will also be understood that each feature

of each embodiment discussed herein, may be used in combination with the features of any other embodiment. Thus, the breadth and scope of the present invention should not be limited by the above-described exemplary embodiments, but should be defined only in accordance with the appended claims and their equivalents.

Claims

1. An apparatus for injecting a molten material comprising: an injector for injecting the molten material; a rail; a first guide and a second guide slideably coupled to the rail, the second guide attached to the injector and spaced from the first guide along the rail; and a bell crank including a first location, a second location, and a third location arranged in a triangle, the bell crank rotateably coupled to the first guide at the first location, the bell crank rotateably coupled to the second guide at the second location such that applying a linear force at the third location in a first direction from the second guide towards the first guide translates the first and second guides in the first direction along the rail.
2. The apparatus according to claim 1, wherein the rail includes two lateral opposite sides and the apparatus further comprising an another bell crank, each of the bell crank and the another bell crank is located on a respective one of the two lateral opposite sides of the rail; the another bell crank including a fourth location, a fifth location, and a sixth location arranged in a triangle, the another bell crank rotateably coupled to the first guide at the fourth location, the another bell crank rotateably coupled to the second guide at the fifth location such that applying a linear force at the sixth location in the first direction translates the first and second guides in the first direction along the rail.
3. The apparatus according to claim 2, wherein at any location along the rail, the first guide is substantially restricted from distancing from the rail in a second direction along a line not substantially parallel to the rail.
4. The apparatus according to claim 3, wherein the rail includes two longitudinal protrusions, one on each respective lateral sides of the rail, the first guide includes a body having two flanges defining a space for partially receiving the rail, each flange adjacent a respective lateral sides of the rail, each flange including a protrusion engaging a respective one of longitudinal protrusions of the rail thereby substantially restricting the first guide from distancing from the rail in the second direction.
5. The apparatus according to claim 4, wherein the bell crank is rotateably coupled to the first guide at the first location via a pin joint and the another bell crank is rotateably coupled to the first guide at the fourth location via a pin joint.
6. The apparatus according to claim 5, wherein the bell crank is rotateably coupled to the second guide at the second location via a pin joint and the another bell crank is rotateably coupled to the second guide at the fifth location via a pin joint.
7. The apparatus according to claim 6, wherein the rail is a linear slide rail.
8. The apparatus according to claim 7, wherein the first guide includes a carriage block.
9. The apparatus of claim 8, wherein the second guide is a trunnion including two pins one on each opposing sides of the trunnion extending away from the rail.
10. The apparatus of claim 9 further comprising a linear actuator rotateably coupled to the bell crank at the third location via a pin joint and rotateably coupled to the another bell crank at the sixth location via pin joint to translate the injector along the rail.
11. The apparatus of claim 10, wherein the linear actuator includes a motor.
12. The apparatus of claim 4 further comprising a third guide slideably coupled to the rail, upstream of the second guide, supporting and attached to the injector.
13. The apparatus of claim 12, wherein the bell crank is rotateably coupled to the first guide at the first location via a pin joint and the another bell crank is rotateably coupled to the first guide at the

fourth location via a pin joint.

14. The apparatus of claim 13, wherein the bell crank is rotateably coupled to the second guide at the second location via a pin joint and the another bell crank is rotateably coupled to the second guide at the fifth location via a pin joint.

15. The apparatus of claim 14, wherein the rail is a linear slide rail.

16. The apparatus of claim 15, wherein the first guide includes a carriage block.

17. The apparatus of claim 16, wherein the second guide is a trunnion including two pins one on each opposing sides of the trunnion extending away from the rail.

18. The apparatus of claim 17 further comprising a linear actuator rotateably coupled to the bell crank at the third location via a pin joint and rotateably coupled to the another bell crank at the sixth location via pin joint to translate the injector along the rail.

19. The apparatus of claim 18, wherein the linear actuator includes a motor.
